MIL-STD-188-340
21 May 1971

MILITARY STANDARD

LONG HAUL COMMUNICATIONS STANDARDS
EQUIPMENT TECHNICAL DESIGN STANDARDS FOR
VOICE ORDERWIRE MULTIPLEX

SLHC

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DEPARTMENT OF DEFENSE
Washington, D.C. 20305

Long Haul Communications Standards
Equipment Technical Design Standards
for Voice Orderwire Multiplex

MIL-STD-188-340

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommend corrections, additions, or deletions should be reported to the preparing activity. (Director, Defense Communications Agency, ATTN: Code, 330, Washington, D.C. 20305).

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FOREWORD

1. This standard specifies the minimum electrical characteristics of orderwire multiplexers used in the Defense Communications System (DCS) below the message baseband.

2. The need for the standard has become apparent as various orderwire configurations have been procured for various subsystems of the DCS which are not fully compatible with each other.

3. This standard limits its application to the multiplexer itself and does not cover the telephone terminal sets, four or six way-four wire bridges, amplifiers, pads, signaling units and other ancillary devices. The standards for these other devices will be specified in other Military Standards.

4. This standard addresses orderwire multiplex which are available without further advances in the state-of-the-art. It is not the intent of the standard to prevent development of new techniques which may accomplish the functions of an orderwire in a more efficient and economical manner. Its basic function is to standardize the electrical performance characteristics of orderwire multiplexers procured by the Military Departments for the DCS.

5. The standards described herein are suitable for normal voice and teletype orderwire use. However, for special applications where data transmission or a complex alarm monitoring channel is involved in subsystem design, or when the voice channel is to be used in a tandem multihop configuration, considerations other than those addressed in this standard may be involved. For specialized use, the channel characteristics specified in *Defense Communications Agency Circular*, DCAC 330-175-1, Defense Communications System (DCS) Engineering-Installation Standards Manual, paragraphs 3.2.5 - 3.2.5.9.1.2 will be more appropriate.

6. The orderwire multiplex subsystem design shall consider the radio equipment characteristics in the 0.3 kHz - 12 kHz region particularly with respect to equipment that is presently in the DCS inventory. These characteristics vary from radio set to radio set and may vary, e.g., by 10 dB in insertion loss from the mission baseband channel characteristics. Special considerations such as impedance, available filtering, limiting and combiner characteristics should be investigated.

NOTE: DCAC 330-175-1 is being replaced by a series of Military Standards. As the new publications are issued, appropriate paragraphs will be superseded, although the remainder of the documents will be still considered valid for use as DCS engineering-installation standards.

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1. SCOPE

1.1 Purpose. The purpose of this standard is to provide the characteristics of orderwire multiplexers to be utilized in the Defense Communications System (DCS) for use below the message baseband. The standards are to be incorporated in specifications for procurement of orderwire multiplexers for subsystems of the DCS.

1.2 Application. This standard applies to all frequency division orderwire multiplexers procured for the DCS for use primarily on wideband radio transmission systems, to provide up to three four-wire voice orderwire channels. These channels will be confined to the frequency spectrum below 12 kHz at the interface with the wideband radio equipment.

1.3 Objective. The objective of this standard is to ensure compatibility among all orderwire multiplexers procured for use in the DCS regardless of the subsystem, or project, for which they are procured.

2. REFERENCED DOCUMENTS

2.1 Military Standards and Specifications.

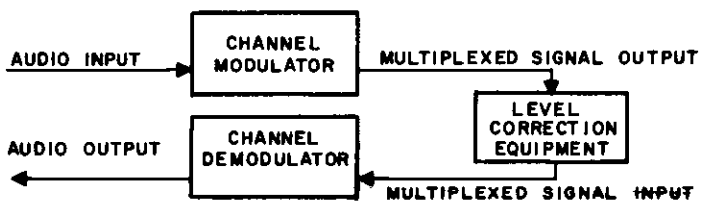
MIL-STD-205	Frequency for Electric Power
MIL-STD-255A	Electric Voltages Alternating and Direct Current
MIL-STD-446A	Environments for Electronic Parts, Tubes and Solid State Devices
MIL-STD-454B	Standard General Requirements for Electronic Equipment
MIL-E-4158E	Military Specification, Electronic Equipment Ground, General Requirements for

2.1.1 Other Documents.

DCAC 330-175-1 Defense Communications System (DCS) Engineering-Installation Standards Manual

3. TERMS AND DEFINITIONS

Back-to-Back Connection. This term refers to a method for connecting equipment. As used in this standard, the output of each channel modulator is connected, through appropriate channel demodulator. This connection is illustrated below for a single channel. All channels have a common connection at the level correction equipment.



Carrier. A sine wave suitable for modulation by the intelligence to be transmitted by the multiplex system.

Carrier Leak. Carrier remaining after carrier suppression in a suppressed carrier transmission system.

Carrier Multiplex. See multiplex, frequency division.

Crosstalk. The phenomenon in which a signal transmitted on one circuit or channel of a transmission system is detectable in another circuit or channel.

Crosstalk, Intelligible. Crosstalk giving rise to intelligible sounds.

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Crosstalk, Near-End. Crosstalk that is propagated in a disturbed channel in the direction opposite to the direction of propagation in the disturbing channel. The terminal of the disturbed channel at which the near-end crosstalk is present is ordinarily near to, or coincides with, the energized terminal of the disturbing channel.

Crosstalk, Unintelligible. Crosstalk giving rise to unintelligible sounds.

dB. Decibel. The standard unit for expressing transmission gain or loss and relative power ratios. The decibel is one tenth the size of a Bel which is too large a unit for convenient use. Both units are expressed in terms of the logarithm to the base 10 of a power ratio, the decibel formula being:

$$dB = 10 \log_{10} \frac{P_1}{P_2}$$

Power ratios may be expressed in terms of voltage or current. If the resistances for both the power measurements are the same, they cancel out in the power ratio so the formulas in terms of voltage or current become as follows:

$$dB = 10 \log \frac{E_1^2}{R_1} \Big/ \frac{E_2^2}{R_2} = 10 \log \frac{I_1^2 \quad R_1}{I_2^2 \quad R_2}$$

$$dB = 10 \log \frac{E_1^2}{E_2^2} = 10 \log \frac{I_1^2}{I_2^2}$$

$$dB = 20 \log \frac{E_1}{E_2} = 20 \log \frac{I_1}{I_2} \quad \text{where } R_1 = R_2$$

dBa. dBrn adjusted. Weighted circuit noise power in dB referred to 3.16 picowatts (-85 dBm), which is 0 dBa. Use of F1A-line or HA1-receiver weighting is indicated in parentheses as required, viz., dBa (F1A).

dBm. dB referred to one milliwatt; employed in communication work as a measure of absolute power values. Zero dBm equals one milliwatt.

dBm0. Absolute power in dBm referred to, or measured at, a point of zero relative transmission level. (The dBm0 figure for a given signal characterizes it wherever the signal may appear elsewhere in the circuit.) See transmission level.

dBrn. dB above reference noise. It is the weighted circuit noise power in dB referred to 1.0 picowatt (-90 dBm), which is 0 dBrn. Use of 144-line, 144-receiver, C-message, or flat weighting is indicated in parentheses as receiver, e.g., dBrn (144-line).

DCS. Defense Communications System.

Distortion, Delay. See distortion, envelope delay.

Distortion, Envelope Delay. Distortion caused by the different transit times in a transmission system for different frequencies. Envelope delay distortion is usually specified as the difference between the maximum and minimum transit time for frequencies within a specified band.

Distortion, Harmonic. The production of harmonic frequencies at the output by the nonlinearity of a circuit when a sinusoidal signal is applied to the input. The amplitude of distortion is usually a function of the amplitude of the input signal.

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Distortion, Intermodulation. Nonlinear distortion characterized by the appearance of frequencies in the output, equal to sums and differences of integral multiples of the component frequencies present in the input.

NOTE: Harmonic components also present in the output are usually not included as part of the intermodulation distortion. When harmonics are included, a statement to that effect should be made.

Distortion, Nonlinear. Distortion caused by a deviation from a linear relationship between the input and output of a system or component.

Erect Position. An erect position of a translated channel is a position in which an increasing signal frequency in the untranslated channel causes an increasing signal frequency in the translated channel.

Express Orderwire. Voice and data communications circuit between selected Technical Control Facilities (TCF's) serving as Facility Coordination Office (FCO, Circuit Control Office (CCO), Special Circuit Control Office (SCCO) or Intermediate Control Office (ICO) for a large number of links, groups, trunks, or circuits.

FDM. Frequency division multiplex.

Guard Band. A frequency band between two channels that gives a margin of safety against mutual interference.

Hertz. Unit of frequency equal to 1 cycle per second.

Impedance. The total opposition offered to the flow of an alternating current. It may consist of any combination of resistance, inductive reactance, and capacitive reactance.

Insertion Loss. The insertion loss of a transmission system (or component thereof) inserted between two impedances, Z_e (transmitter) and Z_r (receiver), is the ratio of the power measured at the receiver, Z_r, before insertion of the transmission system considered, to the power measured after insertion. It is expressed in dB. If the resulting number in dB thus obtained is negative, an insertion gain is indicated.

Inverted Position. An inverted position of a translated channel is a position in which an increasing signal frequency in the untranslated channel causes a decreasing signal frequency in the translated channel.

kHz. Kilohertz. One thousand hertz.

Limiting. A process by which the output of a device is prevented from exceeding a predetermined amplitude.

Linearity. A constant relationship between signal processing devices' input and output characteristics such as frequency, amplitude, phase and time, over a designated range. See Distortion, Nonlinear.

Link Orderwire. Voice and data communications circuits between adjacent communications facilities interconnected by a DCS transmission link.

Local Orderwire. Communications circuit between Technical Control Facility and selected terminal locations within the DCS station complex.

Longitudinal Balance. Electrical symmetry with respect to ground.

Multiplex. Use of a common channel in order to make two or more channels, either by splitting of the frequency band transmitted by a common channel into narrower bands, each of which is used to constitute a distinct channel (frequency division multiplex) or by allotting this common channel, in turn, to constitute different intermittent channels (time division multiplex).

Multiplex, Frequency Division. A multiplex system in which the available transmission frequency is divided into narrower bands, each used for a separate channel.

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Noise, Intermodulation. In a transmission path or device, that noise which is contingent upon modulation and results from any nonlinear characteristic in the path or device.

Noise, Measurement Units. Noise is usually measured in terms of power, either relative or absolute. The decibel is the base unit for these measurements. A suffix is usually added to denote a particular reference base or specific qualities of the measurement. (See Weighting, Noise.) Noise measurement units defined in this standard are dBa, dBa(F1A), dBa(HA1), dBa0, dBm, dBm0, dBm, Psoph., dBm0p, dBrn, dBrn (144-line), dBrn (144-receiver), dBrn (C-Message), dBrn (f_1 - f_2), pW and pWp.

Noise, Quantizing. In a modulation system that employs a quantizing process, quantizing noise is that noise that is caused by the error of approximation. It is an undesirable random distortion signal which is soley dependent on the particular quantization process used and the statistical characteristics of the quantized signal.

Noise Power Ratio, Inband (NPR) (Multichannel Equipment). The noise power ratio, for multichannel equipment, is the ratio of the mean noise power measured in any channel, with all channels loaded with white noise, to the mean noise power measured in the same channel, with all channels by the measured channel loaded with noise.

Noise Power Ratio, In Channel (NPR) (Multichannel Equipment). The noise power ratio, for multichannel equipment, is the ratio of the mean noise power measured in any channel, with no input loading on the channels, to the mean noise power measured in the same channel, with only the measured channel loaded with white noise.

Noise Power Ratio, Single Sideband (NPR) (SSB). NPR (SSB) is the ratio of the mean noise powers measured in the notch filter bandwidth for the notch in and the notch out conditions with total system mean noise power output equal for both conditions.

Noise Weighting. See Weighting, Noise.

Orderwire Multiplex. A multiplex carrier set specifically designed for the purpose of carrying orderwire traffic as opposed to one designed for carrying mission message channels.

Pilot. In a transmission system, a pilot is a signal wave, usually a single frequency, transmitted over the system to indicate or control its characteristics.

Psophometric Weighting. A noise weighting established by the International Consultative Committee for Telephony (CCIF, now CCITT), designated CCITT-1960 weighting for use in a noise measuring set or "Psophometer." The shape of this characteristic is virtually identical to that of F1A weighting. The Psophometer is, however, calibrated with a tone of 800 hertz, 0 dBm, so that the corresponding voltage across 600 ohms produces a reading called 0.775 volt. This introduces a 1 -dB adjustment in the formulas for conversion with dBa.

pWp0. Picowatts, psophometrically weighted, referred to a zero transmission level point.

Return Loss. The return loss at the junction of a transmission line and a terminating impedance is the ratio, expressed in dB, of the reflected wave to the incident wave. More broadly, the return loss is a measure of the dissimilarity between two impedances, being equal to the number of decibels that corresponds to the scaler value of the reciprocal of the reflection coefficient, and hence being expressed by the formula

$$\text{Return Loss} = 20 \log_{10} \left| \frac{Z_1 + Z_2}{Z_1 - Z_2} \right| \text{dB}$$

where Z_1 and Z_2 are the two impedances.

Service Channel. A channel normally provided on a radio set to be used by maintenance personnel in working jointly on one radio link.

Test Tone. A tone sent at a predetermined standard level and frequency through a transmission system to facilitate alignment of the gains and losses of devices in the transmission circuit.

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Translation, Frequency. The transfer en bloc of signals occupying a definite frequency band (such as a channel or group of channels) from one position in the frequency spectrum to another in such a way that the arithmetic frequency difference of signals within the band is unaltered.

Transmission Level. The transmission level (TL) of any point in a transmission system as the mean signal power (in dBm) that should present in operation, relative to the signal power at a zero transmission level point. It is usually measured by applying 0 dBm of 1.0 kHz tone at 0 TLP and measuring the power and hence gain or loss to that receiving TL Point. Thus, a point where a reading of -16 dBm is expected when one milliwatt is applied would be a -16 TLP. The transmission level of a point is a function of system design and is a measure of the gain or loss to the operating signal from 0 TLP. See Zero Transmission Level Point. Since field measurements are usually made either to check that the system is operating properly or to adjust the gain or loss to its design value, it is convenient to determine that value in terms of the reading that would be obtained at a 0 TLP location by the use of the formula:

Measurement referred to 0 Transmission Level (in dBm 0)	=	Actual Measurement TL	-	Relative transmission Level of the test point TL

NOTE: Level and power in a Carrier System. Although levels are more important than actual power to the engineer laying out a telephone circuit, the level assignment must be based upon the mean signal power at each location. The transmission engineer interested in the installation or operation of a carrier system must know the mean power at the various locations. Otherwise, there is a possibility of operating the circuits with either more or less average power than the equipment is designed to transmit.

Upright Position. Same as erect position.

Virtual Carrier. The frequency location that carrier energy would occupy if carrier energy were present.

Zero Transmission Level Point. An arbitrarily chosen point in a circuit to which all relative transmission levels are referred.

4. GENERAL STANDARDS

4.1 Frequency Division Orderwire Multiplex Subsystems. The following standards in chapters 4 and 5 are applicable to subsystem design, equipment performance, and subsystem performance of frequency division orderwire multiplex subsystems of the DCS.

4.2 General Design Requirements.

4.2.1 Construction. All electronic equipment shall meet the requirements of MIL-STD-454B, Standard General Requirements for Electronic Equipment. All electronic equipment shall be of modular construction providing interchangeability between similar modules without requiring custom fitting and incorporating as much "repair by replacement" maintenance as possible without requiring special tools and alignment procedures. The equipment service life shall be at least ten years depending on the particular application.

4.2.2 Power. The orderwire multiplex shall be capable of operating from the power sources specified in Requirement 25 of MIL-STD 454B; MIL-STD- 205, MIL-STD-255 and table II of MIL-E-4158E.

4.2.3 Operational Environmental Conditions. Electronic parts, tubes, and solid state devices utilized in the orderwire multiplex shall meet the requirements of MIL-STD-446A. The entire unit shall meet the following requirements:

a. Ambient temperature: -29° to +52° C

b. Relative humidity: 0 to 95%

c. Elevation: Sea level to 15,000 ft. MSL.

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5. SPECIFIC STANDARDS

5.1 General. The standards addressed in this chapter apply to specific characteristics of the audio channel, carrier generating equipment, modulation plan, and the interface between the orderwire multiplex and the radio equipment. These standards are adequate for orderwire use although orderwire subsystem designers should note that:

a. For special applications on data transmission, external equalizers will be required and

b. Tandem performance will depend on the associated radio equipment and propagation characteristics.

5.2 Audio Channel Characteristics. The standards specified in this paragraph shall be met with the multiplexer connected back-to-back with appropriate level correction.

5.2.1 Audio Frequency Test Tone Levels. Multiplex equipment shall be capable of providing a net gain of 23 dB ±1 dB when measured at the demodulator output relative to the signal level at the corresponding channel modulator input. Thus, a -16 dBm, 1 kHz tone input to the channel modulator should result in a +7 dBm output of the corresponding channel demodulator.

5.2.1.1 Channel Modulator Input. The test tone at the channel modulator input pad shall have a value of 0 dBm and by means of pads the input to the channel modulator shall be adjustable to deliver -16 dBm ±0.5 dB to the modulator input. The modulator shall be designed to carry the operating signal power and peak voltages associated with the specified level.

5.2.1.2 Channel Demodulator Output. The 1 kHz tone at the channel demodulator output shall be +7 dBm when 0 dBm0 is applied at a transmitting 0 TLP, but shall be continuously adjustable in the demodulator or demodulator output from -1 dBm to +10 dbM. The demodulator shall be designed to carry the operating signal power and peak voltages associated with the specified levels.

5.2.2 Audio Frequency Impedance. The impedance at the four-wire audio inputs and outputs shall be 600 ohms balanced to ground with a longitudinal balance of not less than 40 dB. The return loss shall not be less than 20 dB over the frequency band of 300 to 3400 Hz when compared to a nonreactive impedance of 600 ohms.

5.2.3 Insertion Loss Versus Frequency Characteristic. The variation in insertion loss of a channel over the frequency range 300-3400 Hz referred to the insertion loss at 1000 Hz shall not exceed -1.0 dB to +3.5 dB. (Note that a negative change in insertion loss is equivalent to an increase in received channel level and a positive change in insertion loss is equivalent to a decrese in received channel level.)

5.2.4 Envelope Delay Distortion. The envelope delay distortion of a channel shall not exceed 600 microseconds over the 1000 to 3000 Hz portion of the channel.

5.2.5 Noise.

5.2.5.1 Idle Noise. With all audio channel modulator inputs terminated with 600 ohms resistors, and all but one audio channel demodulator output terminated with 600 ohms, unterminated demodulator channel output shall not exceed 100 pW0 (+20 dBrnc0).

5.2.5.2 Loaded Noise. With all save one audio input channels loaded with white noise at an input level equivalent to -5 dBm per channel and the corresponding channels in the demodulator terminated at 600 ohms, the noise measured in the unloaded channel shall not exceed 200 pW0 (+23 dBrnc0).

5.2.6 Intelligible Near-End Crosstalk. The near end crosstalk when measured with a 0 dBm0 test signal within the same channel (transmit pair to receiver pair) in the frequency range from 300 to 3400 Hz shall not exceed -50 dBm0; adjacent channel near-end crosstalk shall not exceed -70 dBm0.

5.2.7 Channel Level Stability. The variation in insertion gain due to variation of environmental characteristics within the limits specified in paragraph 4.2.3 and equipment aging over any 30 day period shall not exceed ±1.0 dB.

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5.2.8 Harmonic Distortion. The total power of the second and thrid harmonic distortion products produced at the four wire receive terminals with a 0 dBm0, 1000 Hz test signal applied to the four wire transmit terminals shall not exceed -34 dBm0.

5.2.9 Frequency Translation Error. Any frequency within the voice channel pass band entering the four wire input to the transmit terminal shall be reconstituted at the four wire output of the receive terminal with an error not exceeding ±2 Hz. This requirement includes frequency translation errors due to variations in environmental characteristics within the limits specified in Section 4 and due to equipment aging over any 30 day period.

5.2.10 Channel Carrier Leak. The channel carrier leak shall not exceed a level 30 dB below the reference transmission level.

5.3 Characteristics of Interface with Radio Equipment, 0.3-12 kHz. The characteristics in this paragraph address the HF side of the orderwire multiplexers which, for the purpose of this document, is the side which interfaces with the radio equipment in the frequency range 0.3-12 kHz. See figure 1). A bridge may be inserted at this interface for special applications providing it does not degrade the performance of the multiplexer as stated herein.

5.3.1 Transmission Levels. With a test tone at a level of 0 dBm0 applied to the transmit (VF input) terminals of a channel, the HF output level of the multiplex shall be adjustable to any level from -10 dBm to -45 dBm. The multiplexer shall be capable of accepting the HF output from the radio at a test tone level of from -10 dBm to -45 dBm. The actual levels used in a practical system will be adjusted to conform to those of the radio equipment used.

5.3.2 Impedances. The impedance at the HF input and output shall be optional for the nominal values of 600 ohms, balanced; 135 ohms, balanced; and 75 ohms, unbalanced. The actual value utilized in a real subsystem will depend upon the radio equipment utilized. (See figure 1).

5.3.3 Spurious Products. The level of any product falling in the band of frequencies outside of the 0.3-12 kHz range on the HF side of the multiplexer shall not exceed -60 dBm0.

5.4 Equipment Organization and Modulation Plan.

The equipment shall provide for 1, 2 or 3 channels in the 0.3 kHz to 12 kHz frequency band. The HF line sides (input and output) of the three channels shall be combined internally to form a common HF input and HF output. No further translation will be made at the input to the radio modulator or after output from the radio receiver.

Channel one shall be the direct, untranslated channel in the 0.3 kHz to 3.4 kHz frequency band.

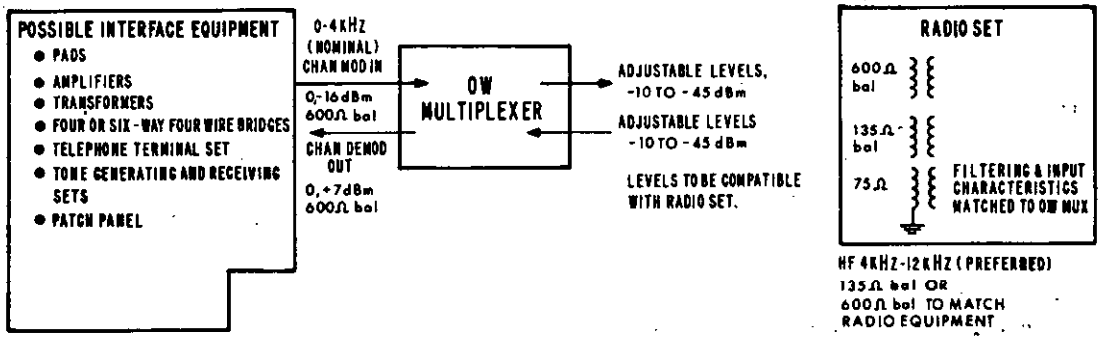


FIGURE 1. CHARACTERISTICS OF INTERFACE WITH RADIO EQUIPMENT, 0.3-12 kHz

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Channels two and three shall be derived as lower sideband, suppressed carrier frequency division multiplexed channels in the 4 kHz and 8 kHz frequency bands, respectively. (See figure 2).

5.5 Carrier Generation. The channel carrier for the channel allocated between 4-8 shall be 8 kHz. The channel carrier for the channel allocated between 8 kHz - 12 kHz shall be 12 kHz.

5.6 Synchronization.

If synchronization of channel carrier oscillators is necessary to meet the requirements of 5.2.9, the synchronizing pilot shall be transmitted on either channel carrier frequency at a level of -16 dBm0. If both of the translated channels are in use, it shall be possible by strapping to transmit either or both the 8 kHz and 12 kHz pilots to the HF line equipment. Oscillators shall be capable of operating in the master or slave mode.

If synchronization is not required to meet the requirements of 5.2.9, the capability of transmitting synchronization pilots, selectable by strapping, shall be provided in order to insure compatibility with other equipment as described in above paragraph.

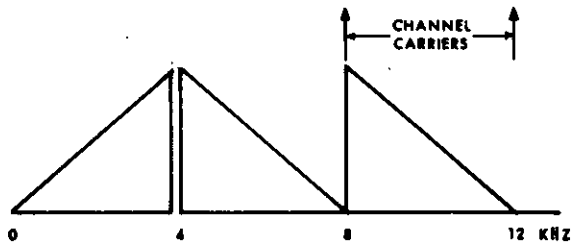


FIGURE 2. MODULATION PLAN FOR ORDER WIRE MULTIPLEXER

CUSTODIANS: ARMY - SC
NAVY - EC
AIR FORCE - 17

PREPARING ACTIVITY:
DCA - DC

(PROJECT SLHC 0040)

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